



UNIVERSITÀ DEGLI STUDI DI SALERNO



UNIVERSITÀ DEGLI STUDI DI SALERNO
DIPARTIMENTO DI INFORMATICA
DIPARTIMENTO DI ECCELLENZA

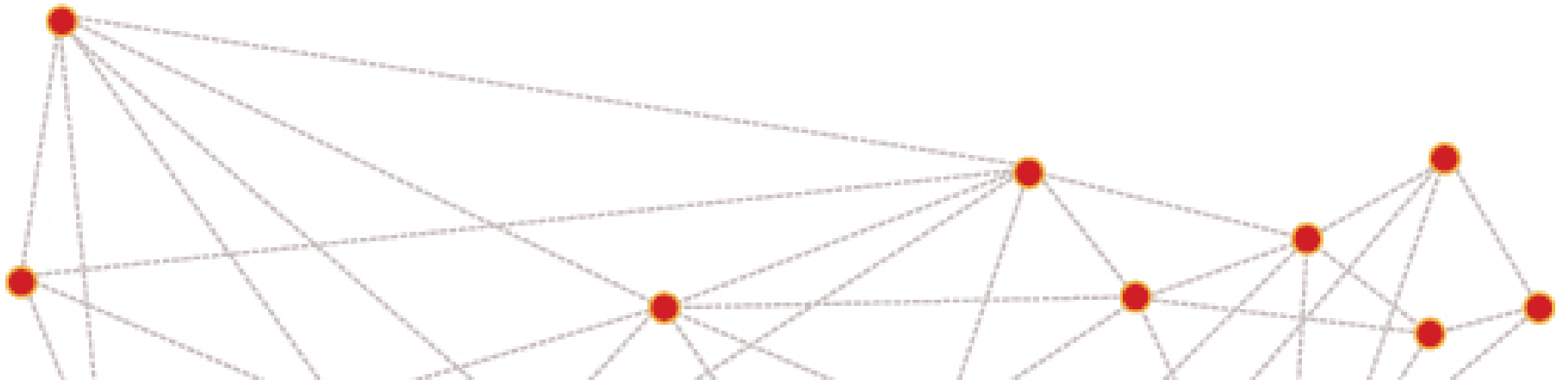


Prof. Carmen Bisogni

C.A.S.A. Course

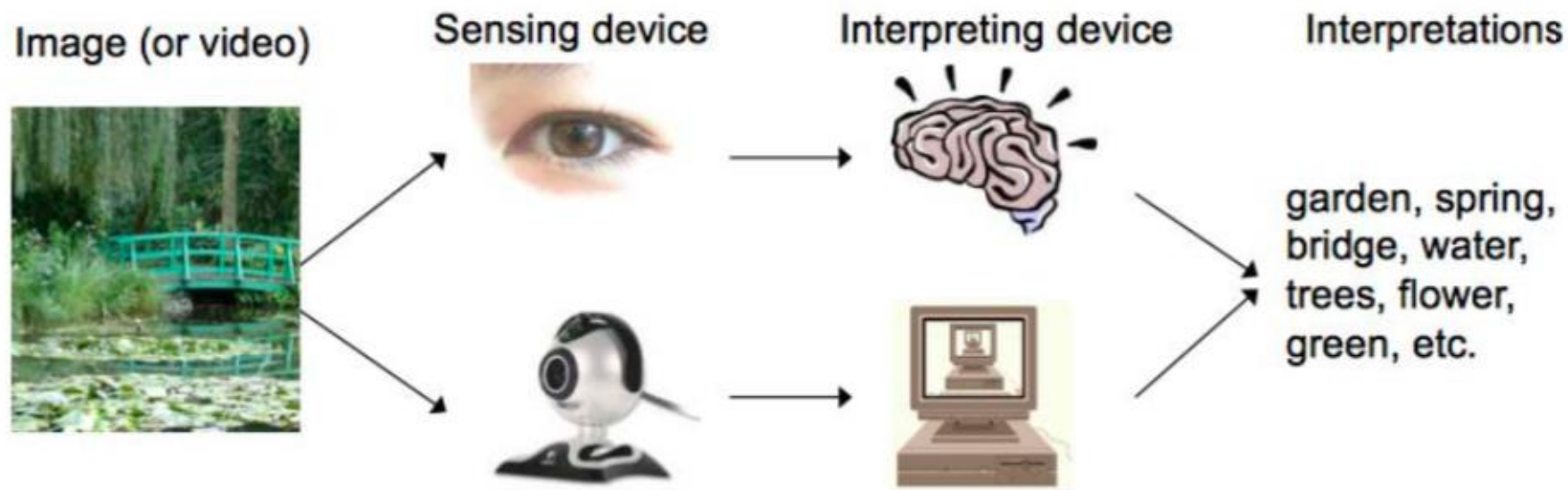
CONTEXT AWARE SECURITY ANALYTICS IN COMPUTER VISION

1





Introduction to Human vs Computer Vision



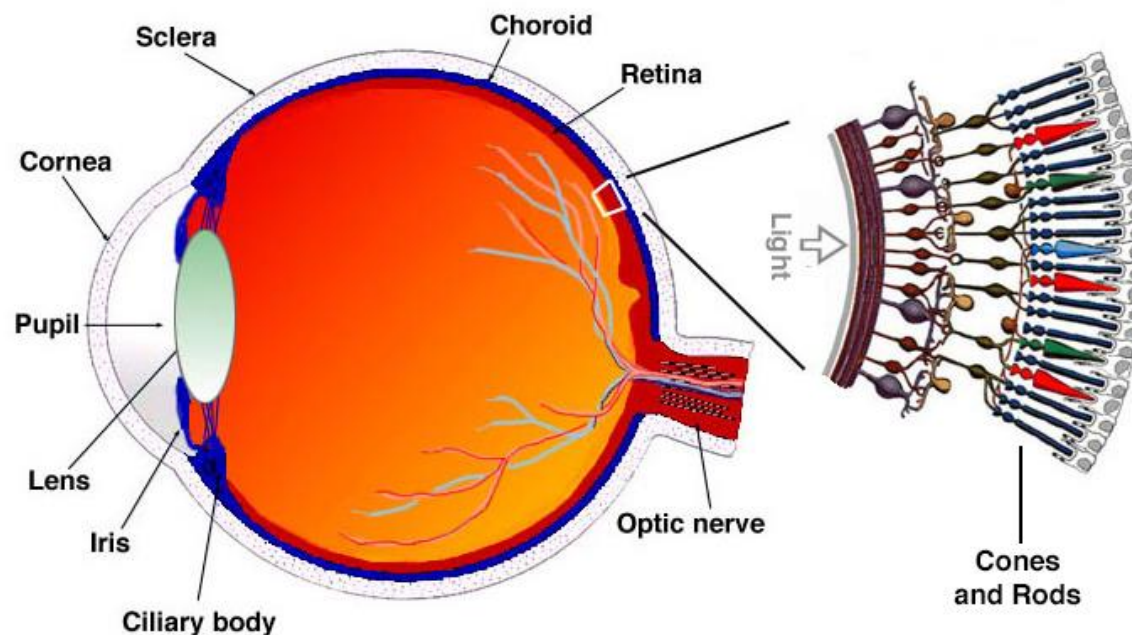


Introduction to Human vs Computer Vision

Left Side: Cross-section of the Eye

The image represents the eye as a sphere with various essential components:

- 1. Cornea** – The transparent front part of the eye that helps focus light.
- 2. Pupil** – The opening at the center of the iris that regulates the amount of light entering the eye.
- 3. Iris** – The colored part of the eye that controls the pupil's size.
- 4. Lens (Crystalline Lens)** – A biconvex lens that changes shape to focus light onto the retina.
- 5. Ciliary Body** – A structure that helps adjust the shape of the lens for focusing.
- 6. Sclera** – The tough, white outer layer of the eye.
- 7. Choroid** – A layer rich in blood vessels that provides oxygen and nutrients to the retina.
- 8. Retina** – The light-sensitive tissue at the back of the eye.
- 9. Optic Nerve** – Transmits visual signals from the retina to the brain.



Human Eye



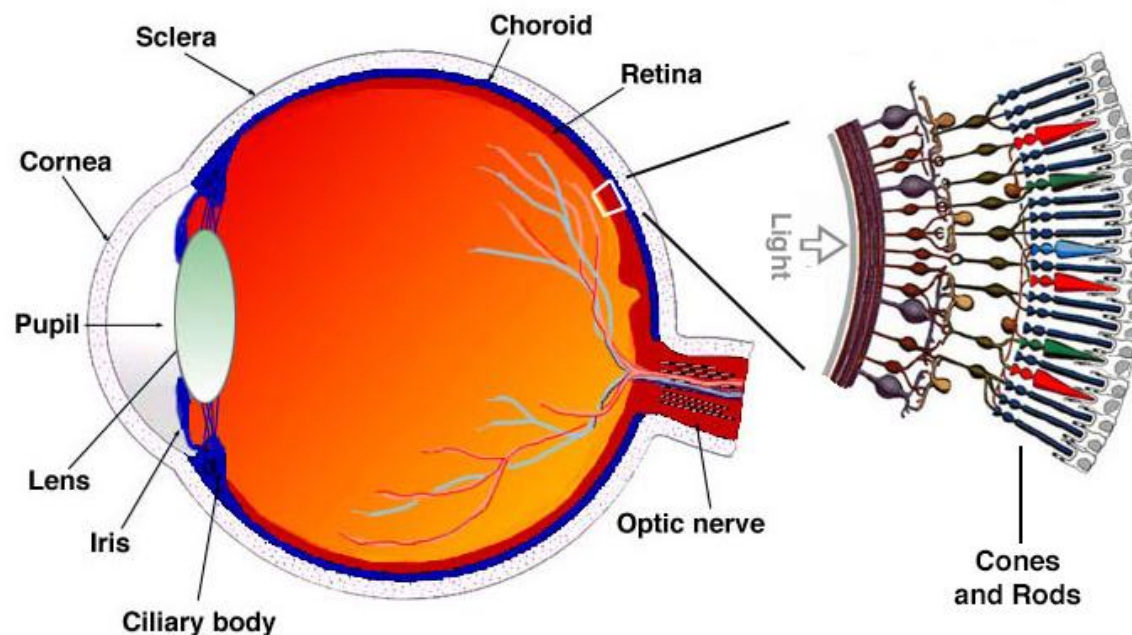


Introduction to Human vs Computer Vision

Right Side: Detailed View of the Retina

The magnified image on the right shows a section of the retina and the photoreceptors responsible for vision:

- **Cones** – Specialized for color vision and detail perception in bright light conditions.
- **Rods** – Sensitive to low light, responsible for night vision and motion detection.
- **Direction of Light** – An arrow indicates how light travels through the retinal layers before being converted into electrical signals.



Human Eye



Introduction to Human vs Computer Vision



How the Human Eye Sees

1. Light Entry

1. Light enters through the **cornea**, which bends and directs it toward the **pupil**.
2. The **iris** regulates the amount of light entering by adjusting the pupil's diameter.
3. Light passes through the **lens**, which changes shape (accommodation) to focus on objects near or far.

2. Image Formation on the Retina

1. Light travels through the **vitreous humor** and reaches the **retina**, *where it is converted into electrical signals by **cones** and **rods**.*
2. **Cones** are responsible for color vision (red, green, and blue), while **rods** are more sensitive to dim light and provide black-and-white vision.

3. Transmission to the Brain

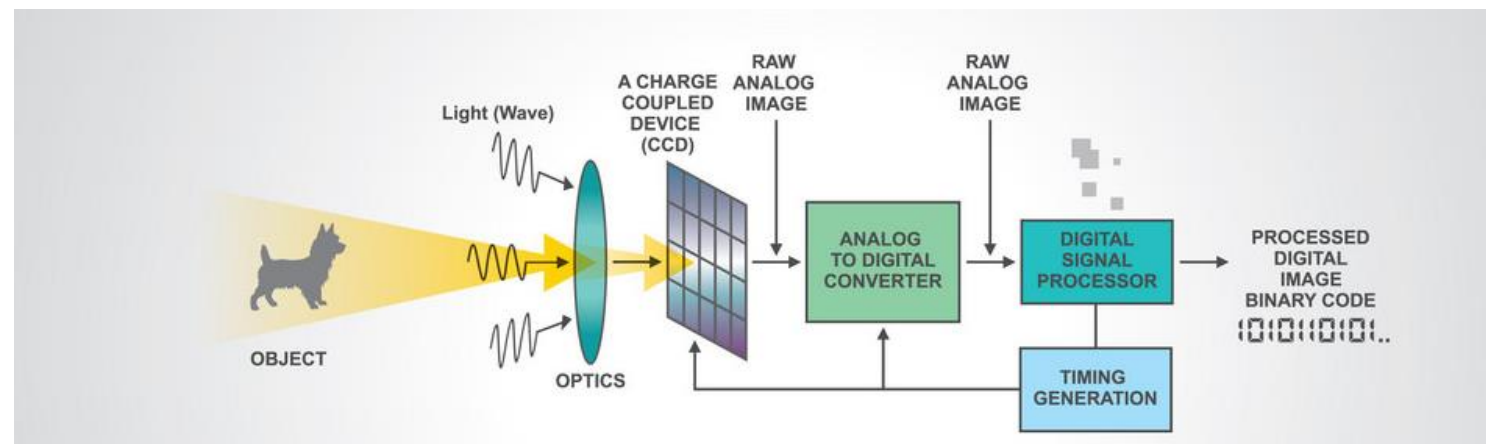
1. Electrical signals are transmitted through the **optic nerve** to the **brain**, specifically to the **visual cortex** in the occipital lobe.
2. The brain processes the received information and combines it to form a clear and detailed image of the surrounding environment.



Introduction to Human vs Computer Vision



Digital Camera



When you press the button to take a photograph with a digital camera, an aperture opens at the front of the camera and light streams in through the lens. There is a piece of electronic equipment that captures the incoming light rays and turns them into electrical signals. This light detector is one of two types, either a charge-coupled device (CCD) or a CMOS image sensor.



Introduction to Human vs Computer Vision



Image Capture: Photoreceptors vs. Sensor

Human Eye:

- The **retina** contains two types of photoreceptors:
 - **Cones** (for color and detail in bright light)
 - **Rods** (for low-light and motion detection)
- The photoreceptors **convert light into electrical signals**, which are processed by the optic nerve and sent to the brain.

Camera:

- The **image sensor** (CMOS or CCD) acts like the retina, detecting light and converting it into electrical signals.
- Instead of cones and rods, the sensor consists of millions of **photosites (pixels)** that detect different levels of brightness and color.
- Bayer filters** (red, green, and blue filters over pixels) allow cameras to capture color, similar to how cones in the eye detect different wavelengths of light.



Introduction to Human vs Computer Vision



We see



Computer sees

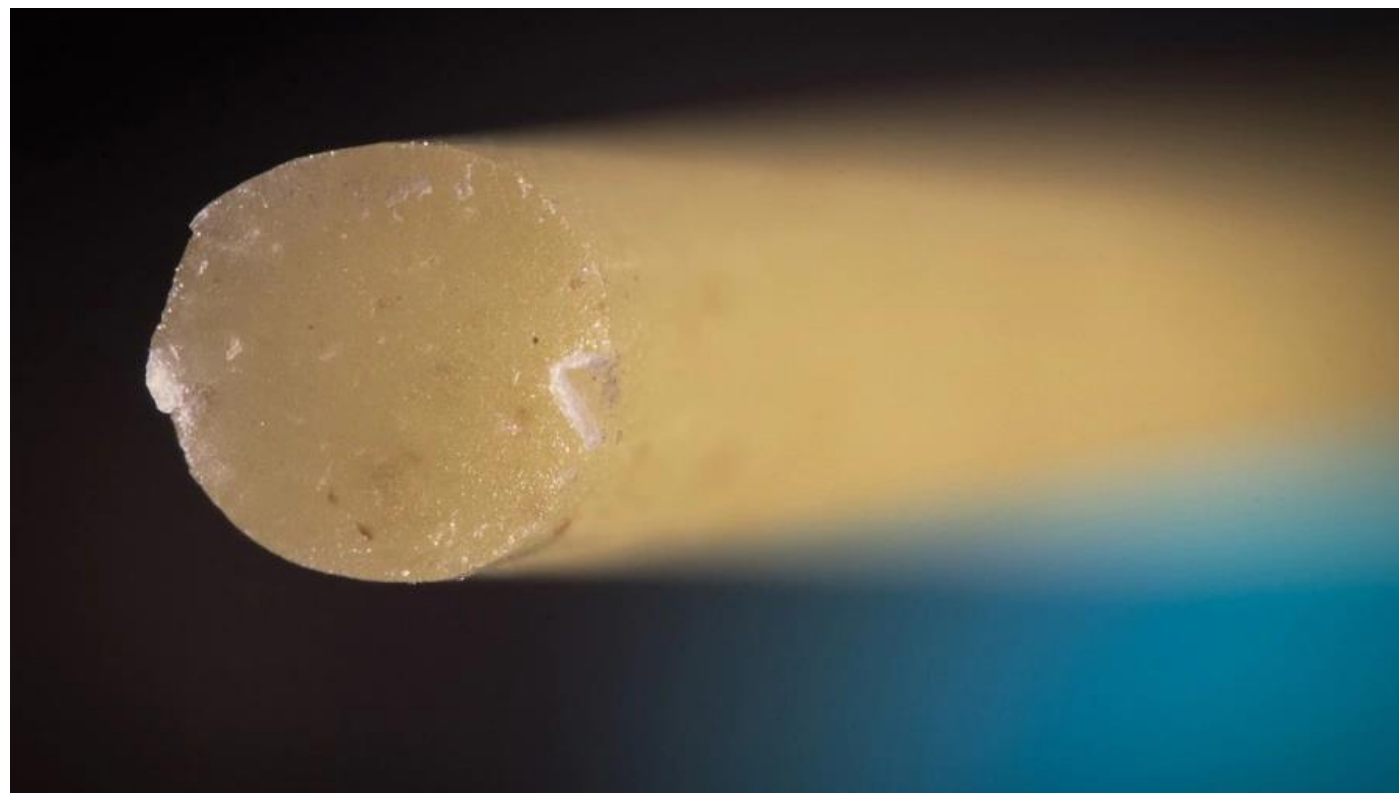




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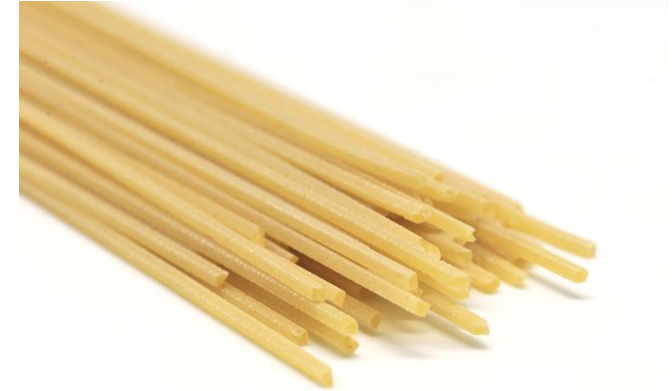


What is this?





Introduction to Human vs Computer Vision



Perspective is of great importance in human vision.



Introduction to Human vs Computer Vision

Not only perspective is of great interest in computer vision but also:

Occlusions



Deformation



Illumination



Intraclass variation





Introduction to Human vs Computer Vision



Learning algorithms, as well as human, need to be trained on a large amount of samples per class.

These are
spaghetti...



But these too...





Introduction to Human vs Computer Vision



But when images are complex, what can we use as training data?

What would you use to
understand which television
broadcast is this?





Introduction to Human vs Computer Vision



People?



«Hell's kitchen»



«McDonald's»
spot



Introduction to Human vs Computer Vision



Clothes?



football match





Introduction to Human vs Computer Vision



Logo!



«Masterchef»



Introduction to Human vs Computer Vision



What band is this?

No logos...





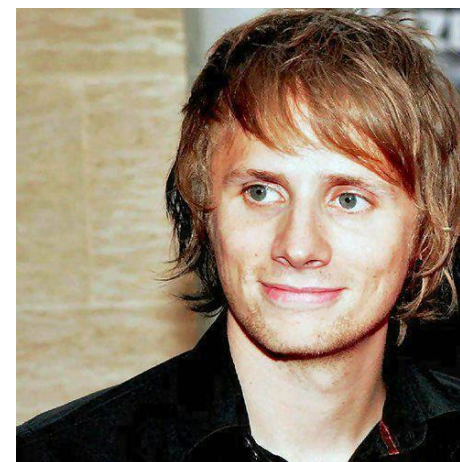
Introduction to Human vs Computer Vision



Face recognition



Dominic Howard
«Muse»





Introduction to Human vs Computer Vision



How do we know if she won?





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Because of prize?





Introduction to Human vs Computer Vision



Other «Masterchef» prizes...



Quite different, isn't it?



Introduction to Human vs Computer Vision



A plate in a museum?





Introduction to Human vs Computer Vision



Facial Expression recognition



Smile! Probably she won...



Computer Vision



«Computer vision is concerned with the automatic extraction, analysis and understanding of useful information from a single image or a sequence of images. It involves the development of a theoretical and algorithmic basis to achieve automatic visual understanding.»

The British Machine Vision Association and Society for Pattern Recognition Retrieved February 20, 2017



Computer Vision: Progress in the Hardware



Kinect

Infrared camera



Drones/Mobile cameras

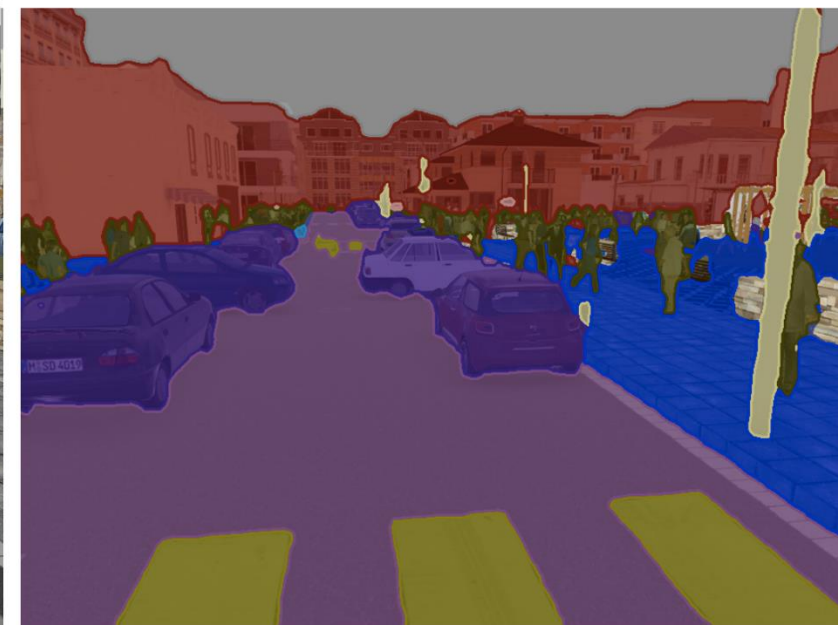




Computer Vision: Progress in algorithms



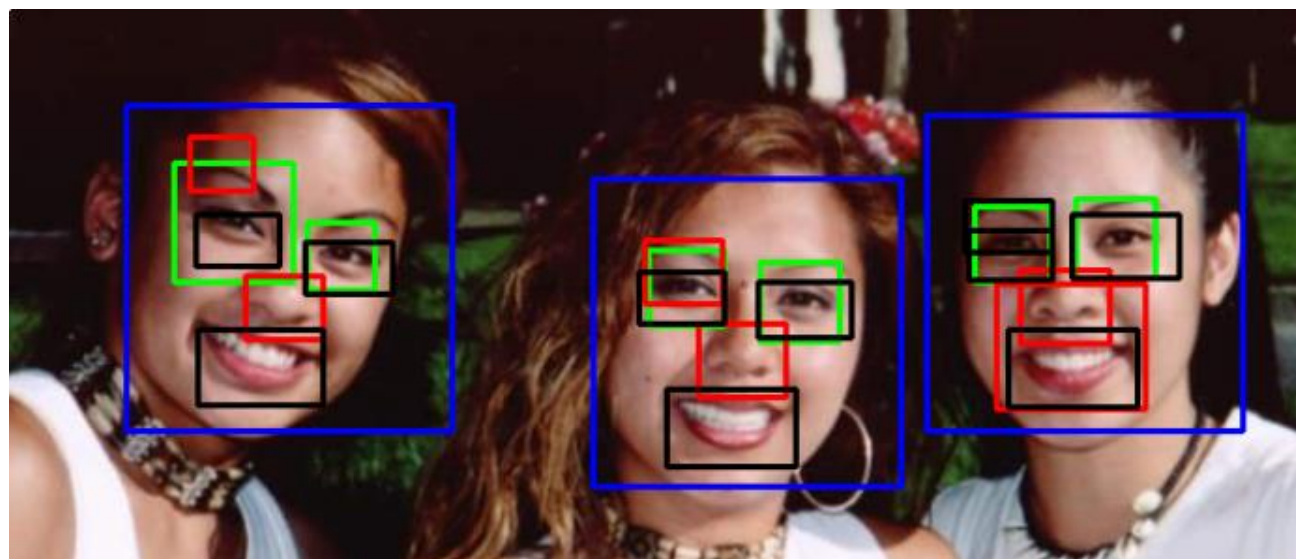
Image
segmentation



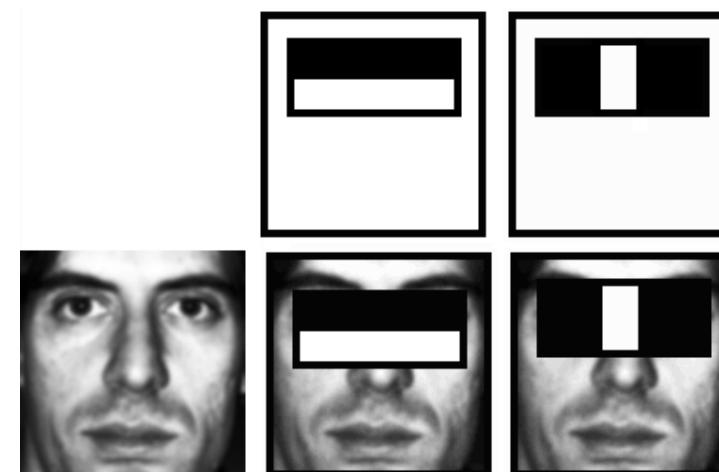
■ Sky ■ Building ■ Road ■ Sidewalk ■ Fence ■ Vegetation ■ Pole ■ Car ■ Sign ■ Pedestrian ■ Cyclist



Computer Vision: Progress in algorithms



Viola-Jones:
Face detection and part of a
face detection

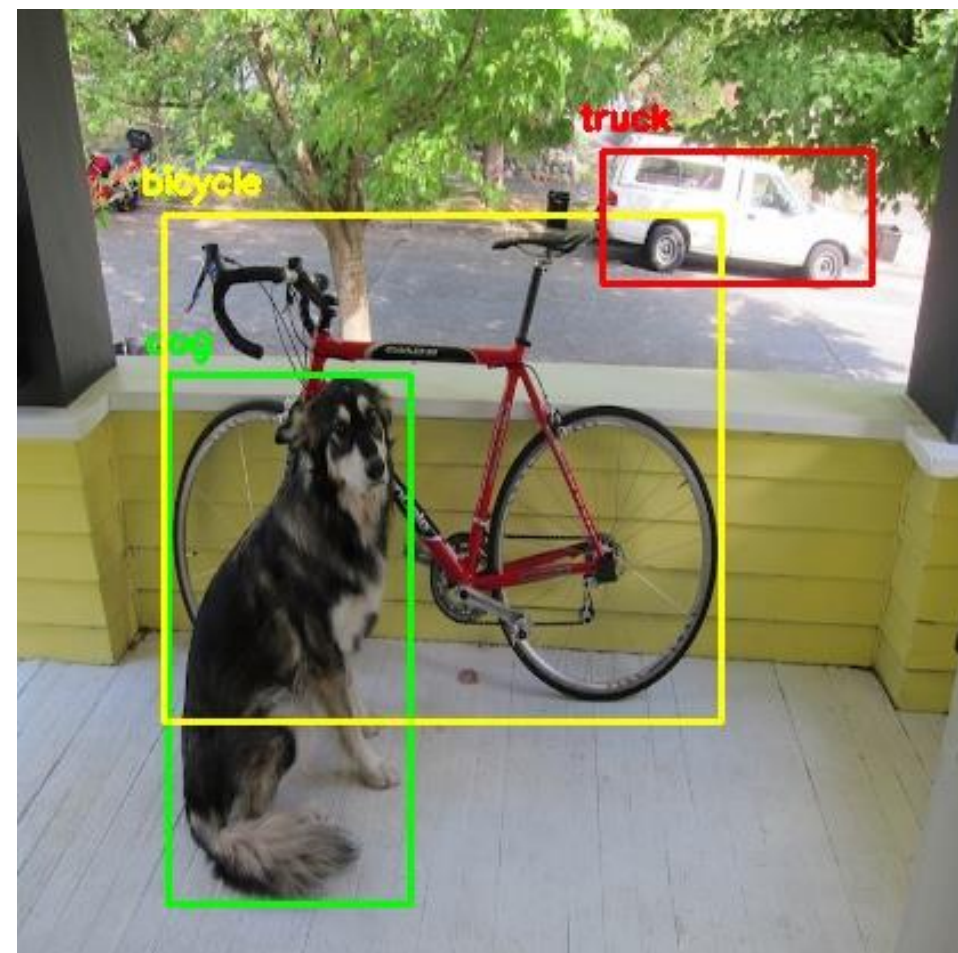
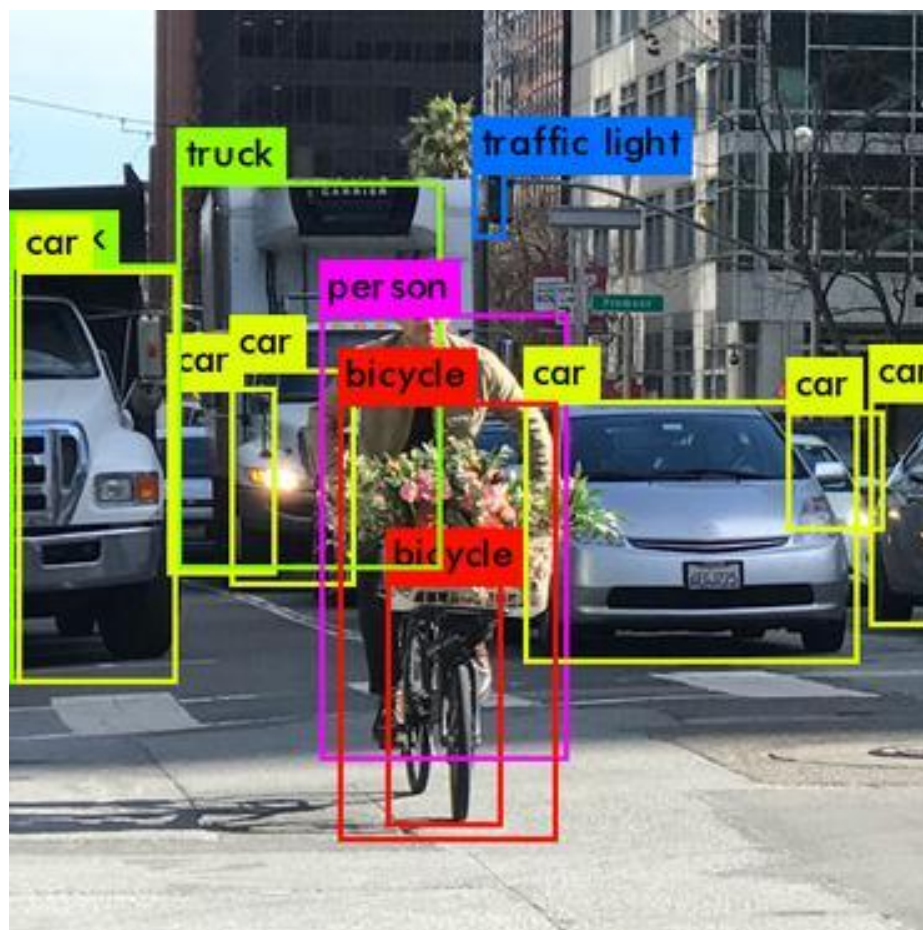




Computer Vision: Progress in algorithms



Object detection





Computer Vision: Progress in algorithms



People/objects tracking...



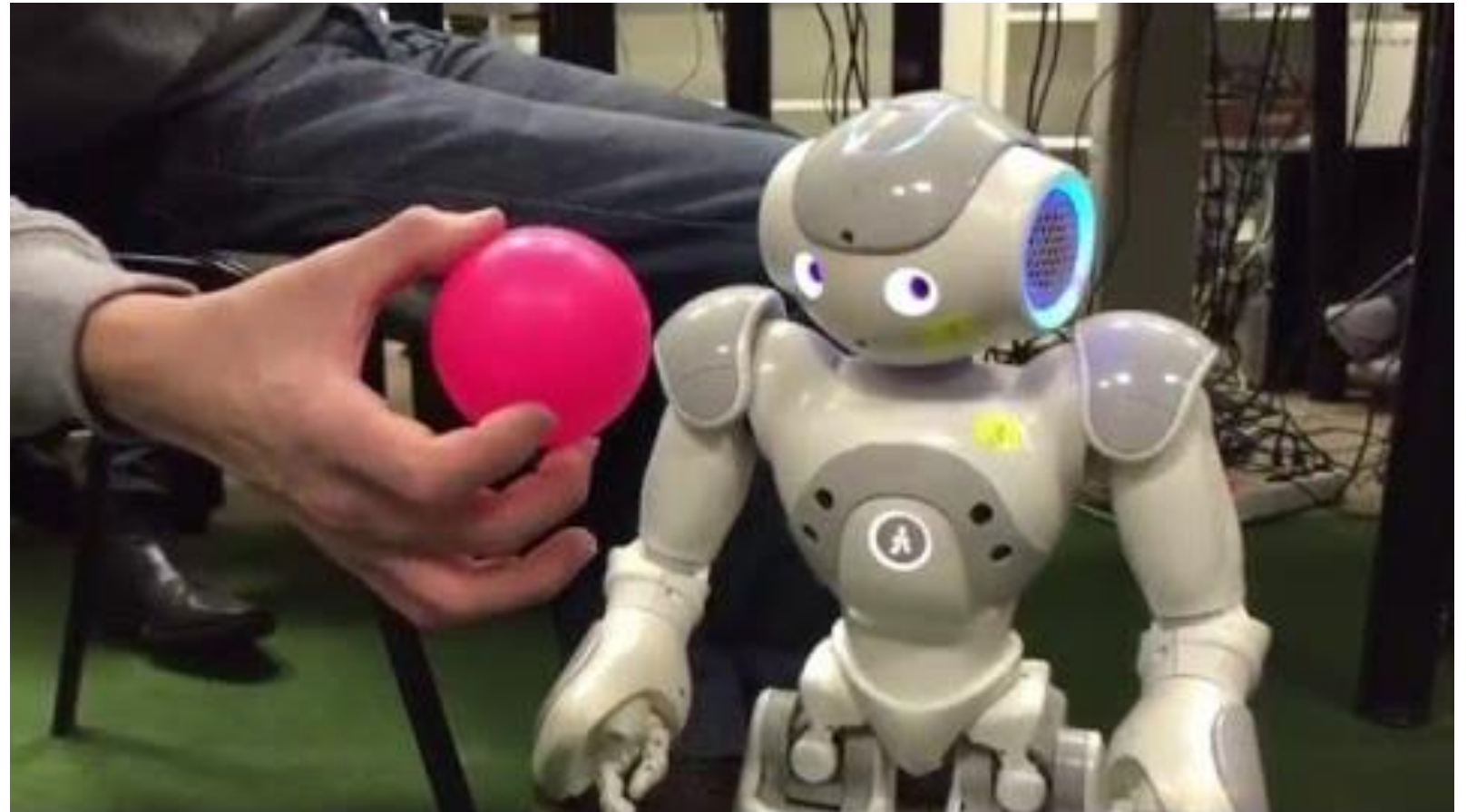
https://www.youtube.com/watch?time_continue=11&v=YhW7YRJvQpg



Computer Vision: Progress in algorithms



...and an example
of application.





Computer Vision: Progress in algorithms



Automatic scene
description



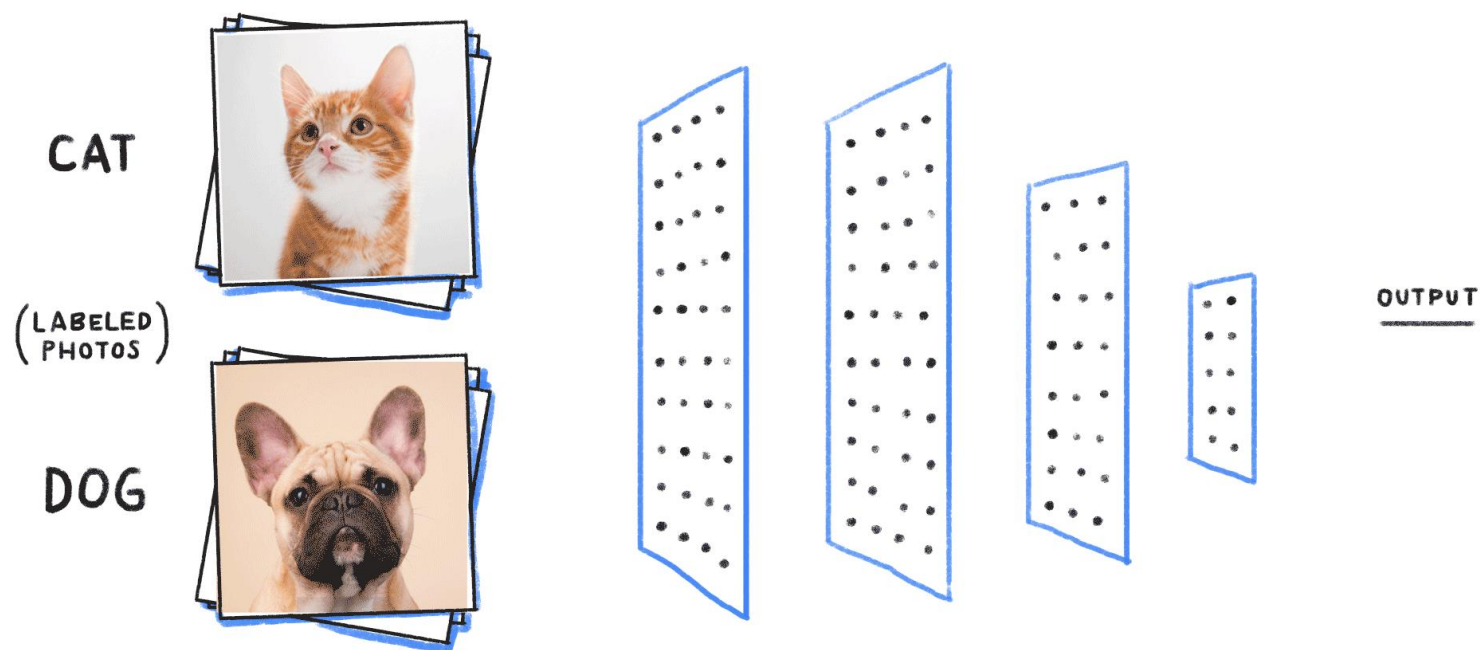
<https://www.youtube.com/watch?v=dKNu9y92iOQ>



Computer Vision: Classification



We are interested mainly in classification algorithms...

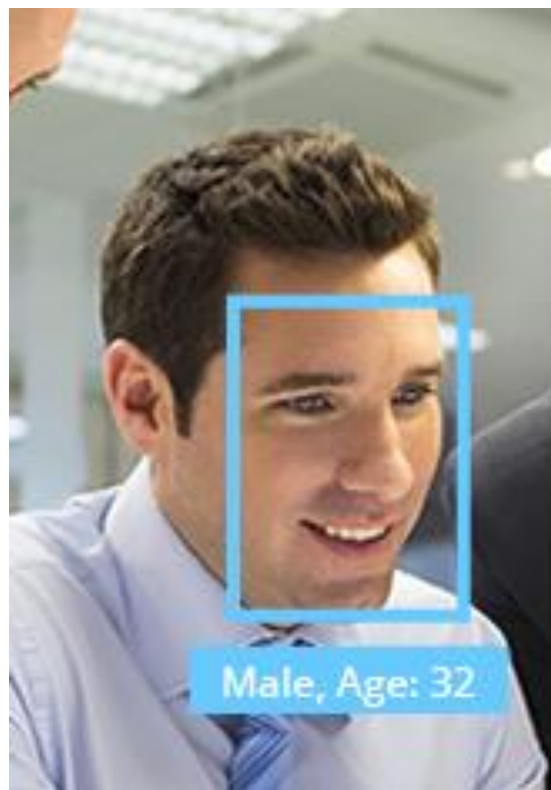




Computer Vision: Classification



...to perform age&gender detection...





Computer Vision: Classification



...emotion classification...



(a) angry



(b) disappointed



(c) laughing



(d) surprised



Computer Vision: Classification



...emotion classification...



(a) angry



(b) disappointed



(c) laughing



(d) surprised



Computer Vision: Classification



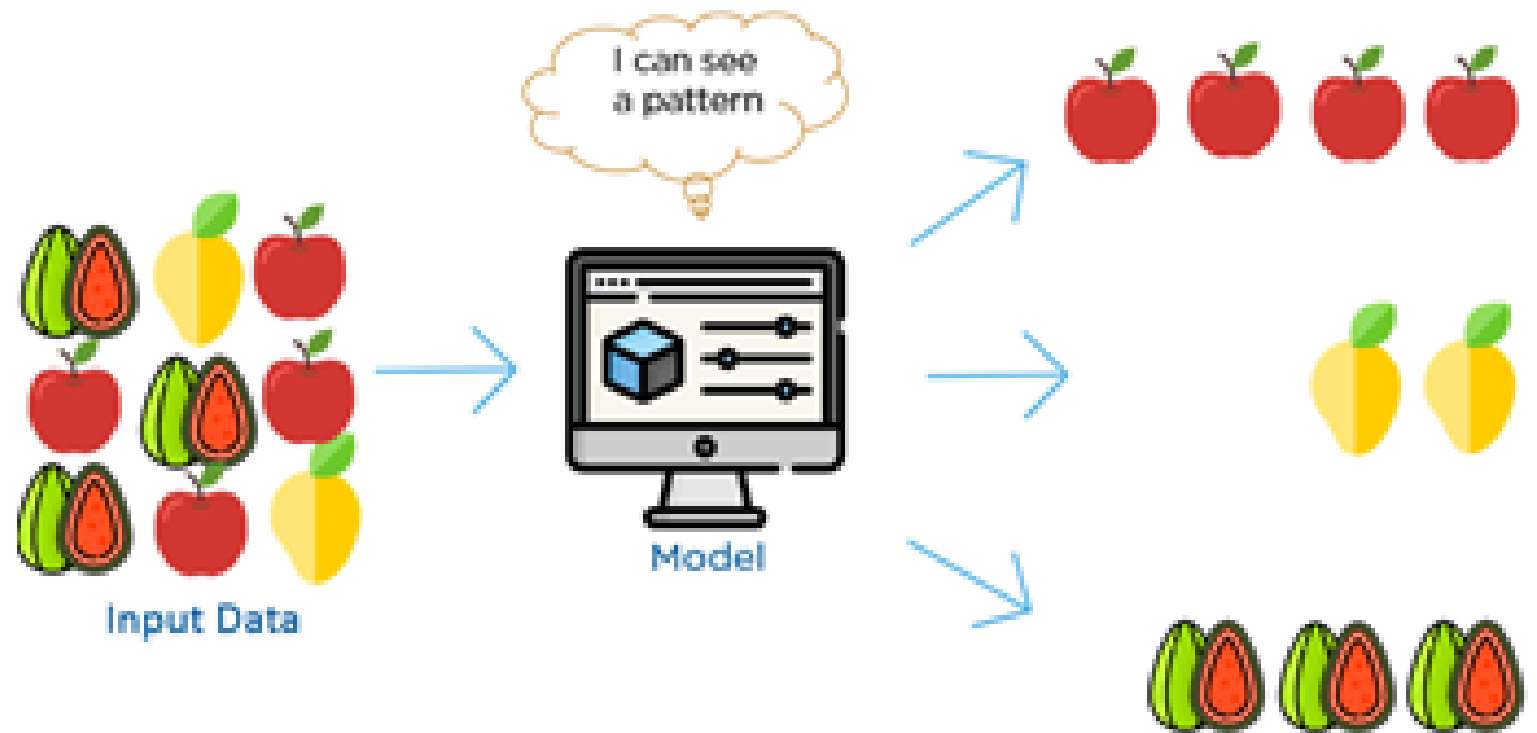
...action recognition
...and so on!





Computer Vision

We are also interested in clusterization*...



*usually when we don't know what inputs are.



Computer Vision



...and progressive models.



*usually used in video and time-dependent data.

<https://www.youtube.com/watch?v=aryhiDMbhT8>

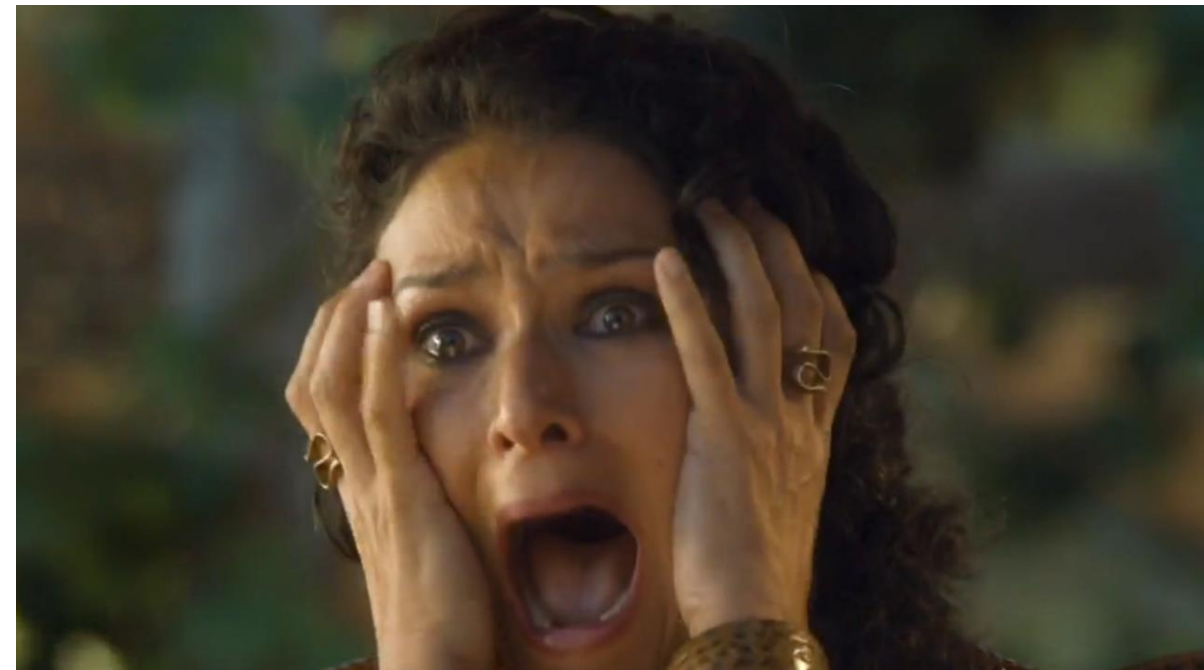


Computer Vision



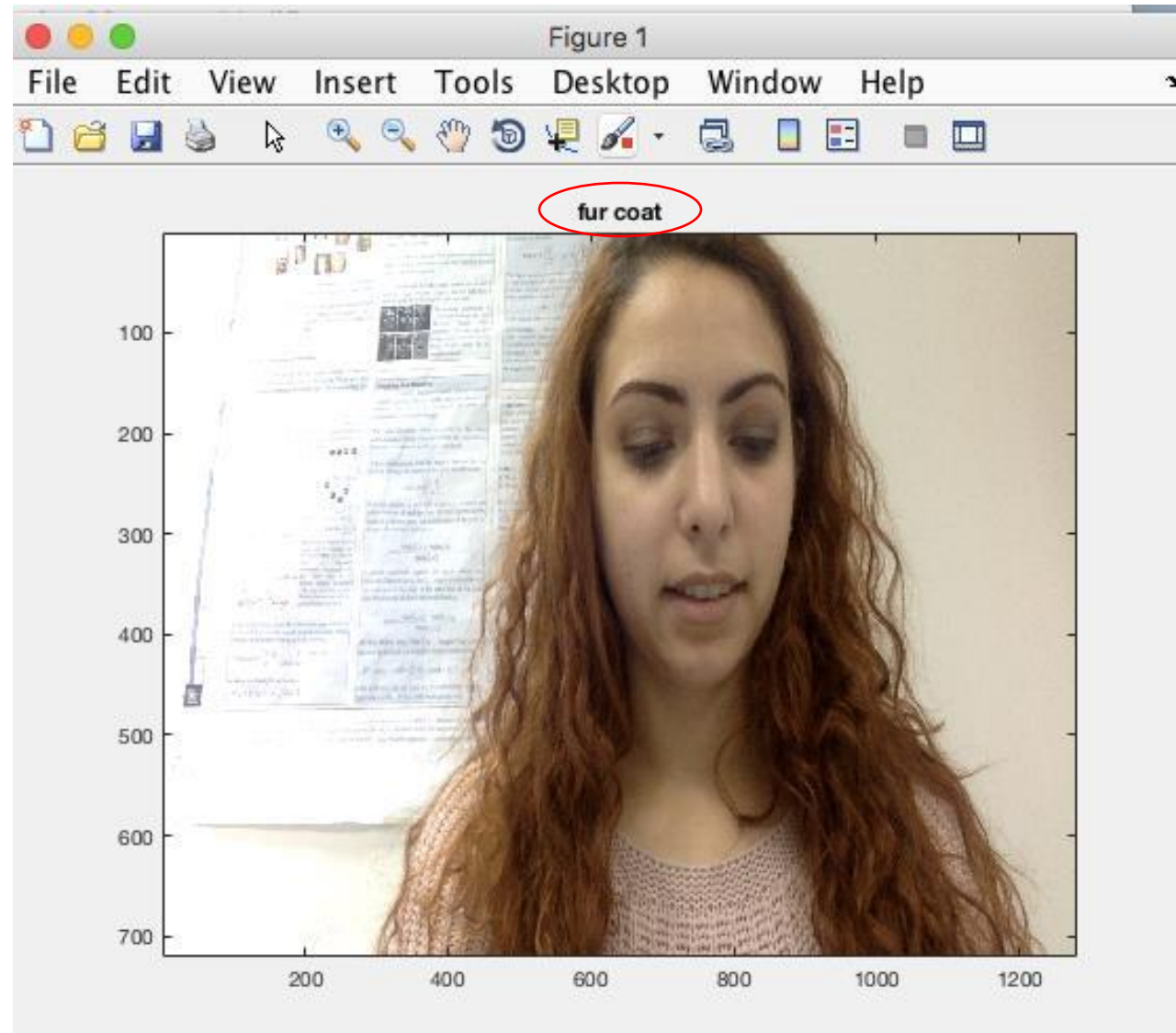
We will perform those tasks with different kind of machine learning algorithms, especially Neural Networks (NN) we will introduce in next lessons.

But, even if we use powerful algorithms as NN, sometimes something goes wrong...





Computer Vision



...I have copper hair...

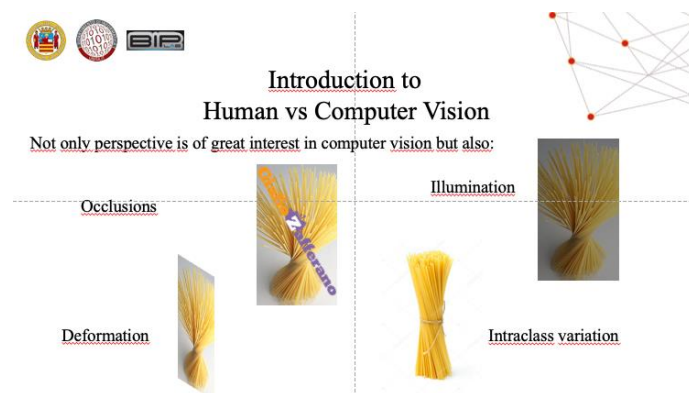
...but I'm definitely not a fur coat!



Computer Vision

What would cause problems like this?

Manipulation of images...as we saw in slide 8.

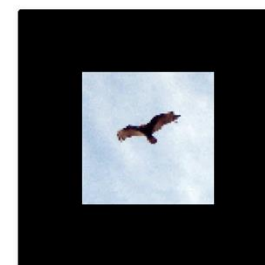


"panda"

Adversarial Noise

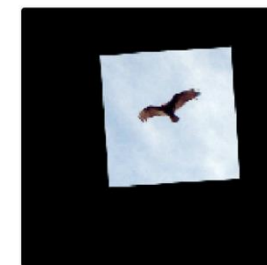
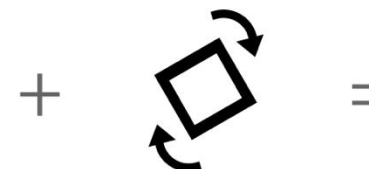


"gibbon"



"vulture"

Adversarial Rotation



"orangutan"



"not hotdog"

Adversarial Photographer



"hotdog"





Computer Vision



...but it is vitally important collecting a lot of sample per class in order to ensure the proper functioning of the neural network!

Interclass variations are essential.



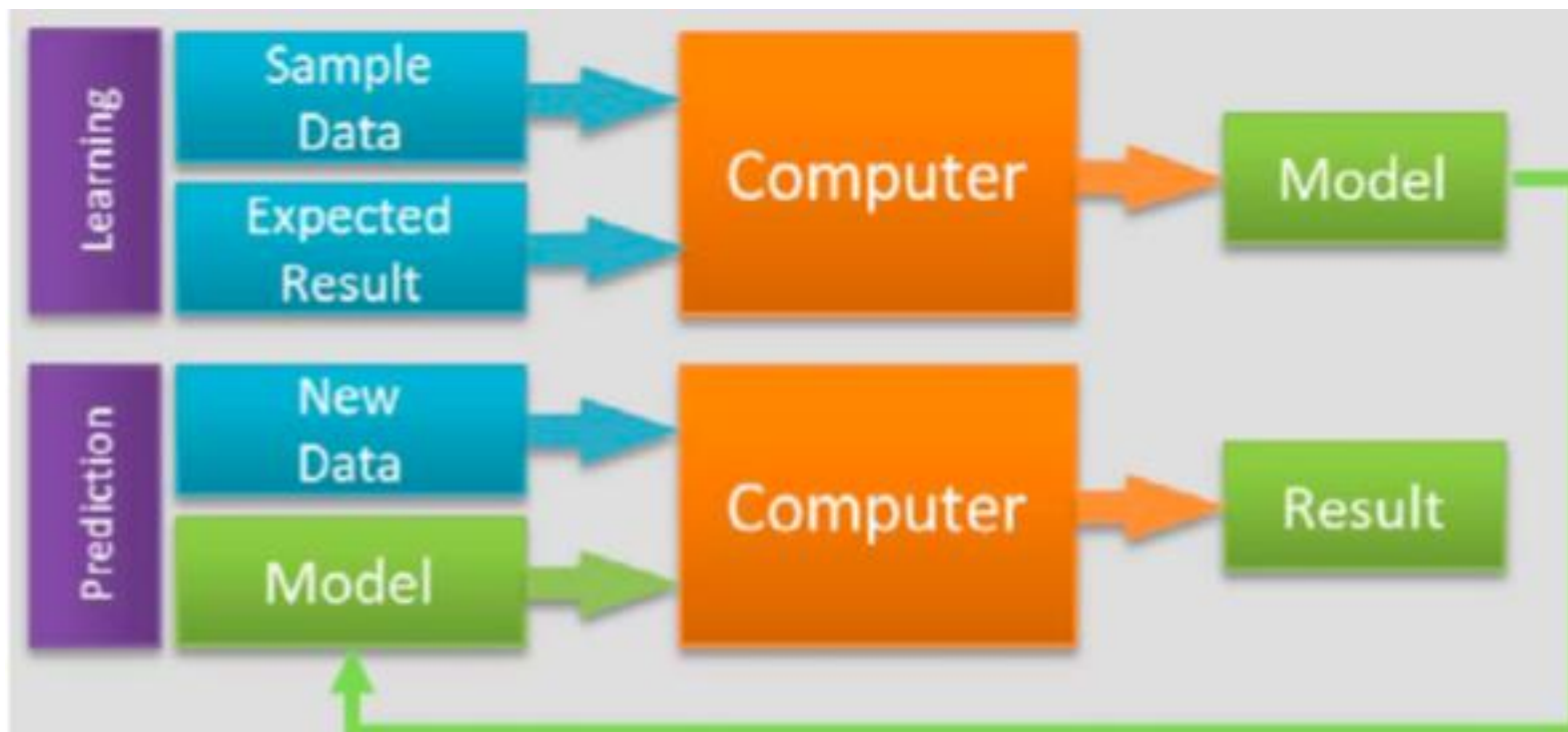


Computer Vision: Traditional models





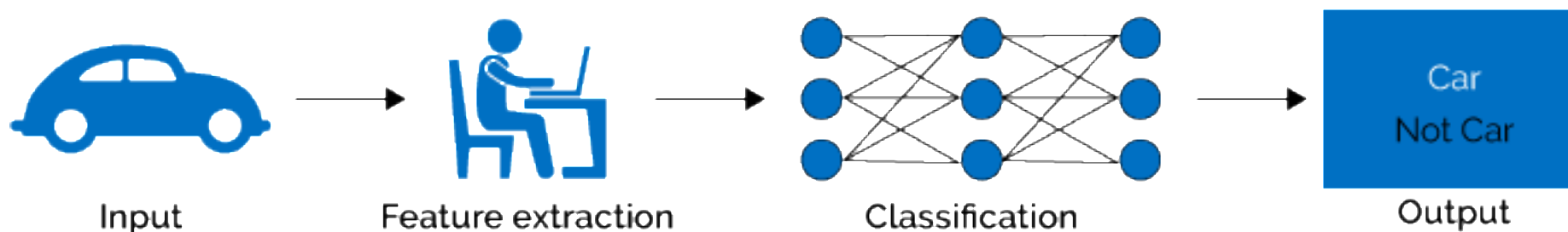
Computer Vision: Deep Learning



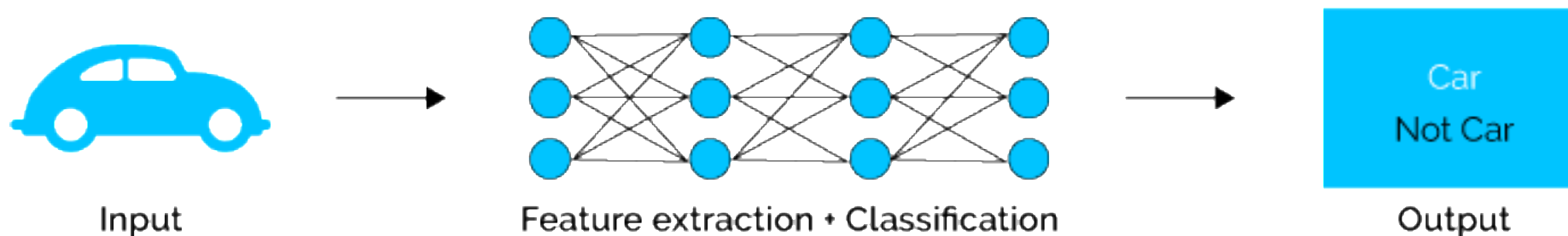


Computer Vision: Machine Learning vs Deep Learning

Machine Learning

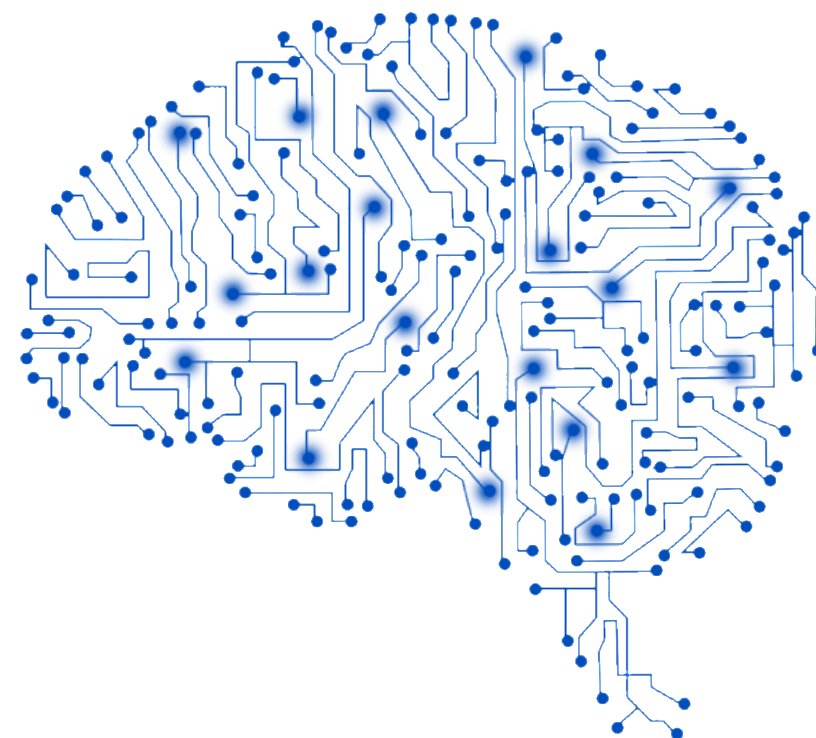
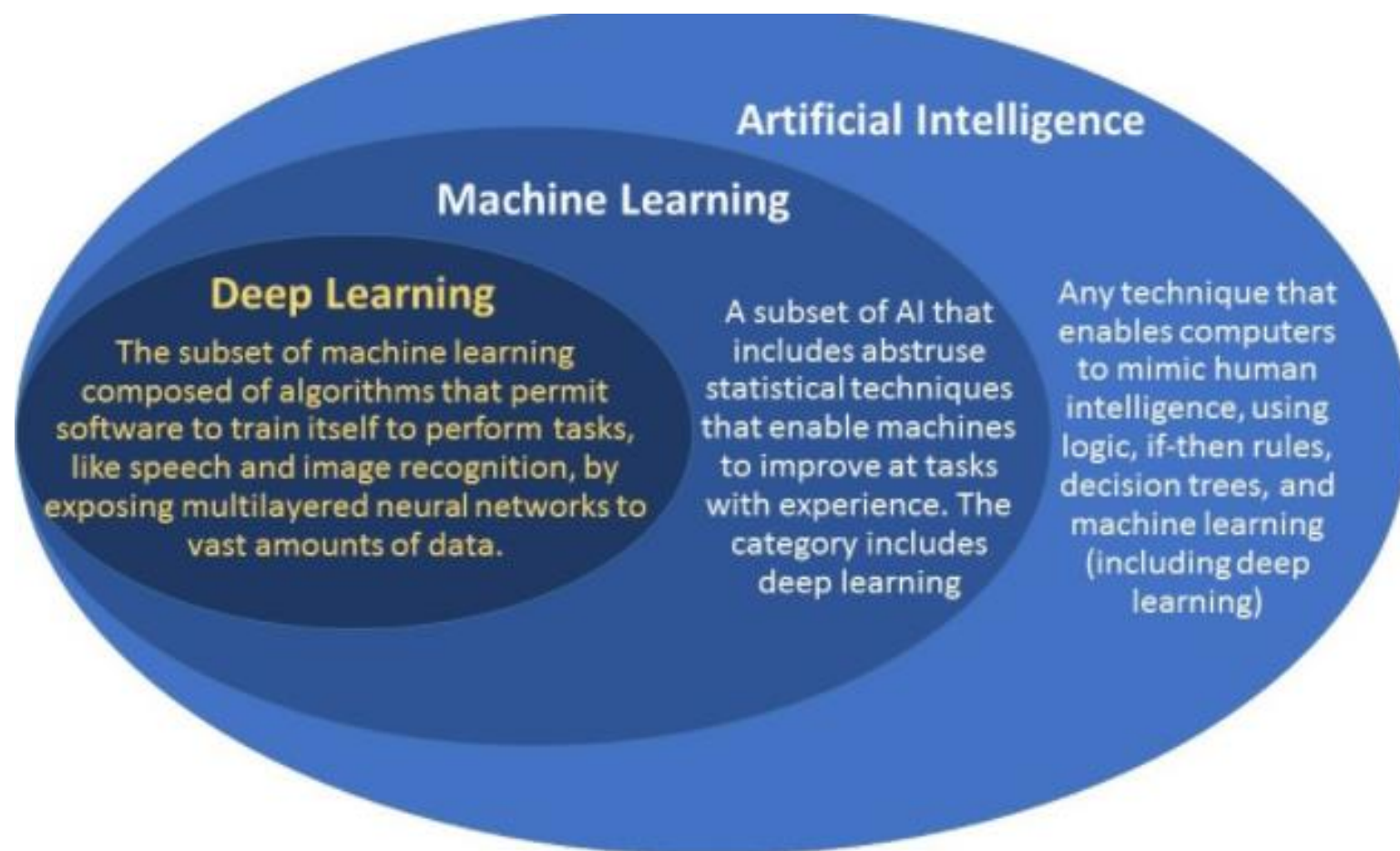


Deep Learning



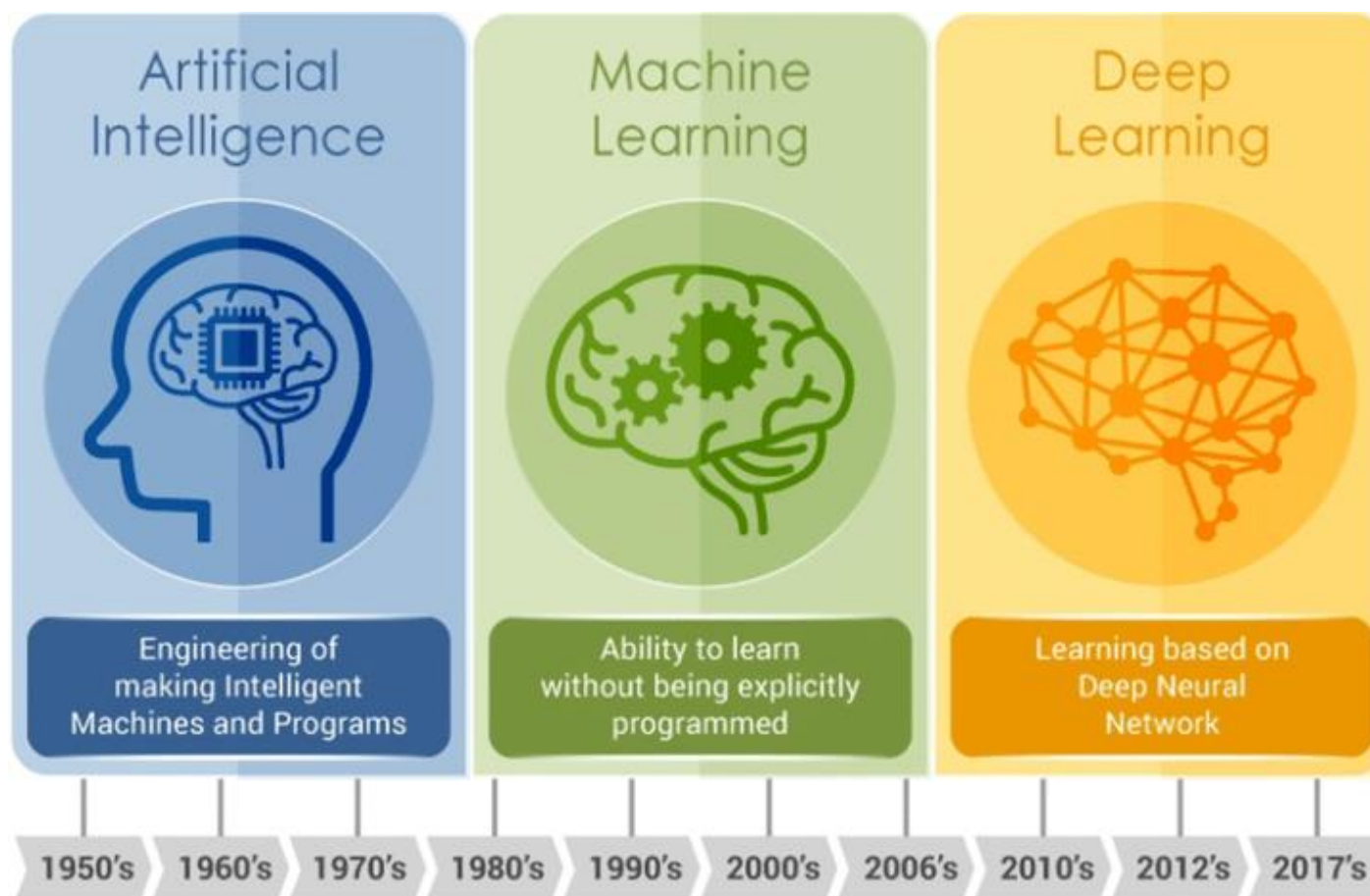


Computer Vision: AI, Machine Learning, Deep Learning





Computer Vision: AI, Machine Learning, Deep Learning

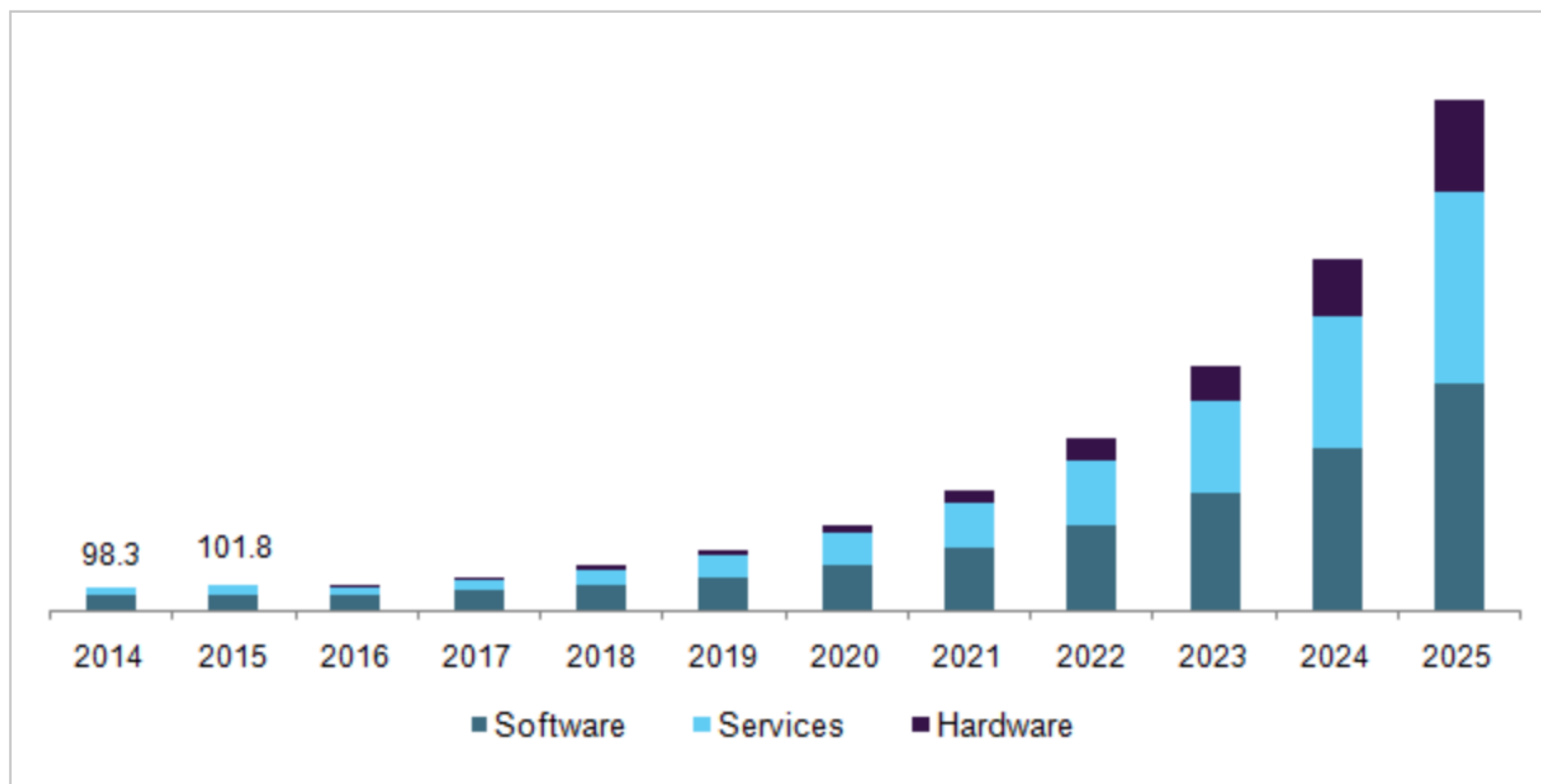




Computer Vision: AI, Machine Learning, Deep Learning



U.S. deep learning market, by solution, 2014 - 2025 (USD Million)







Where are we now? Autonomous Driving





Where are we now? Violence detection





Where are we now? Style Transfer





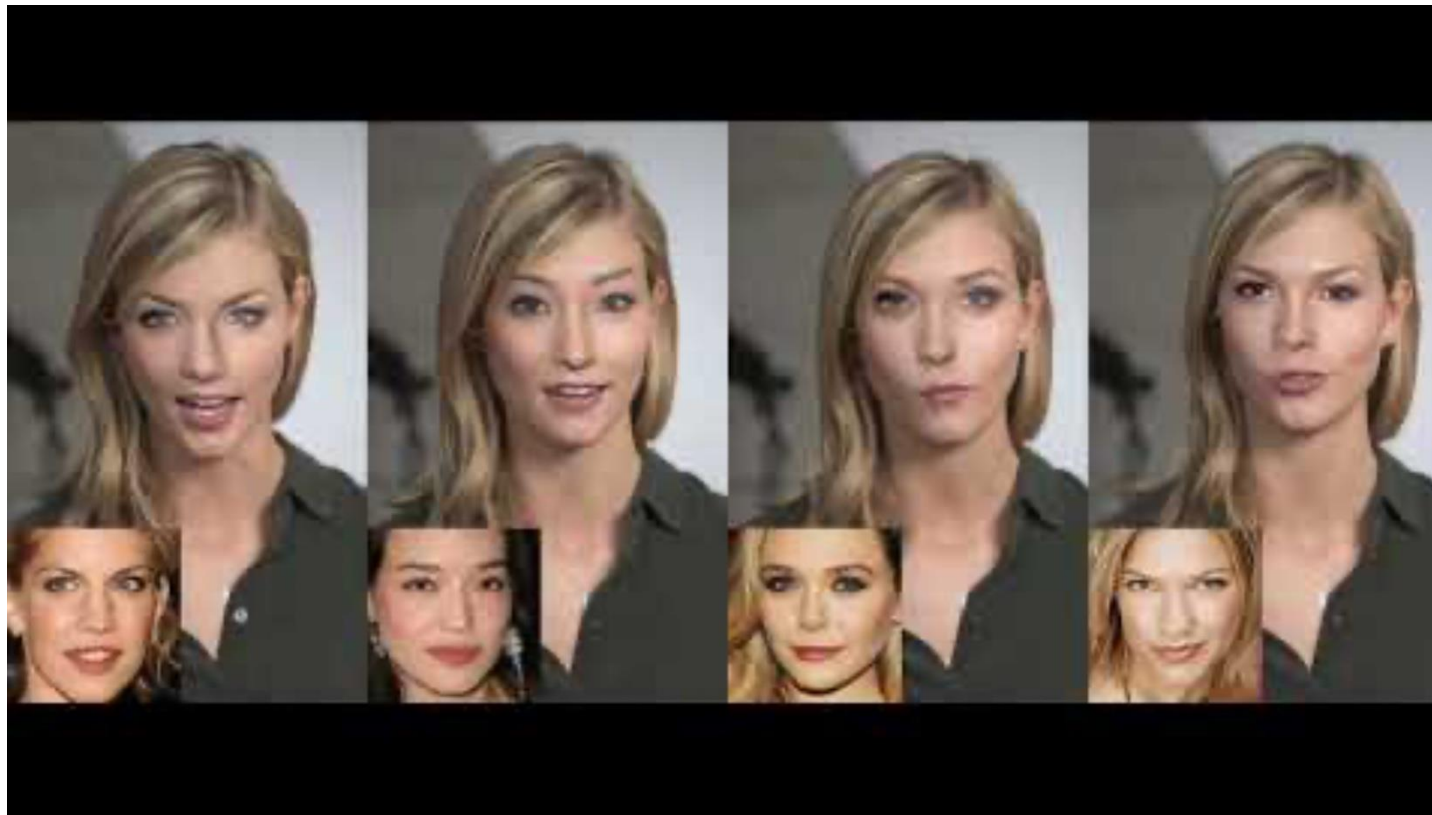
Where are we now? DeepFake - LipSync





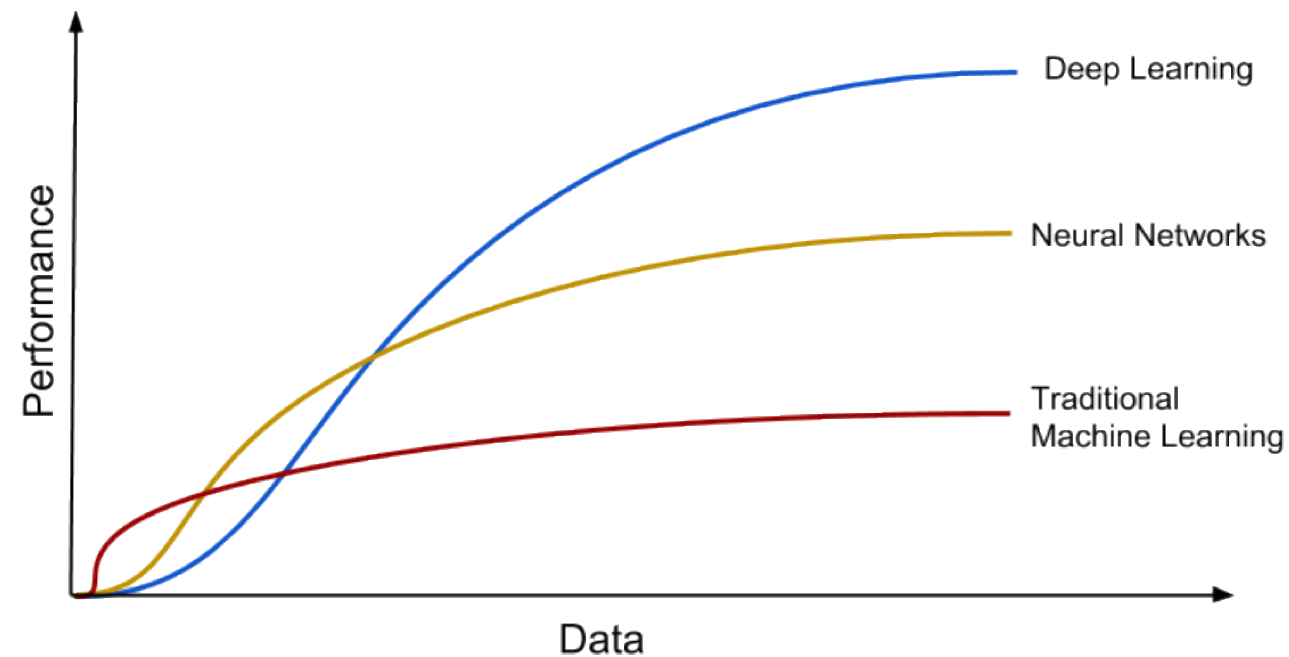
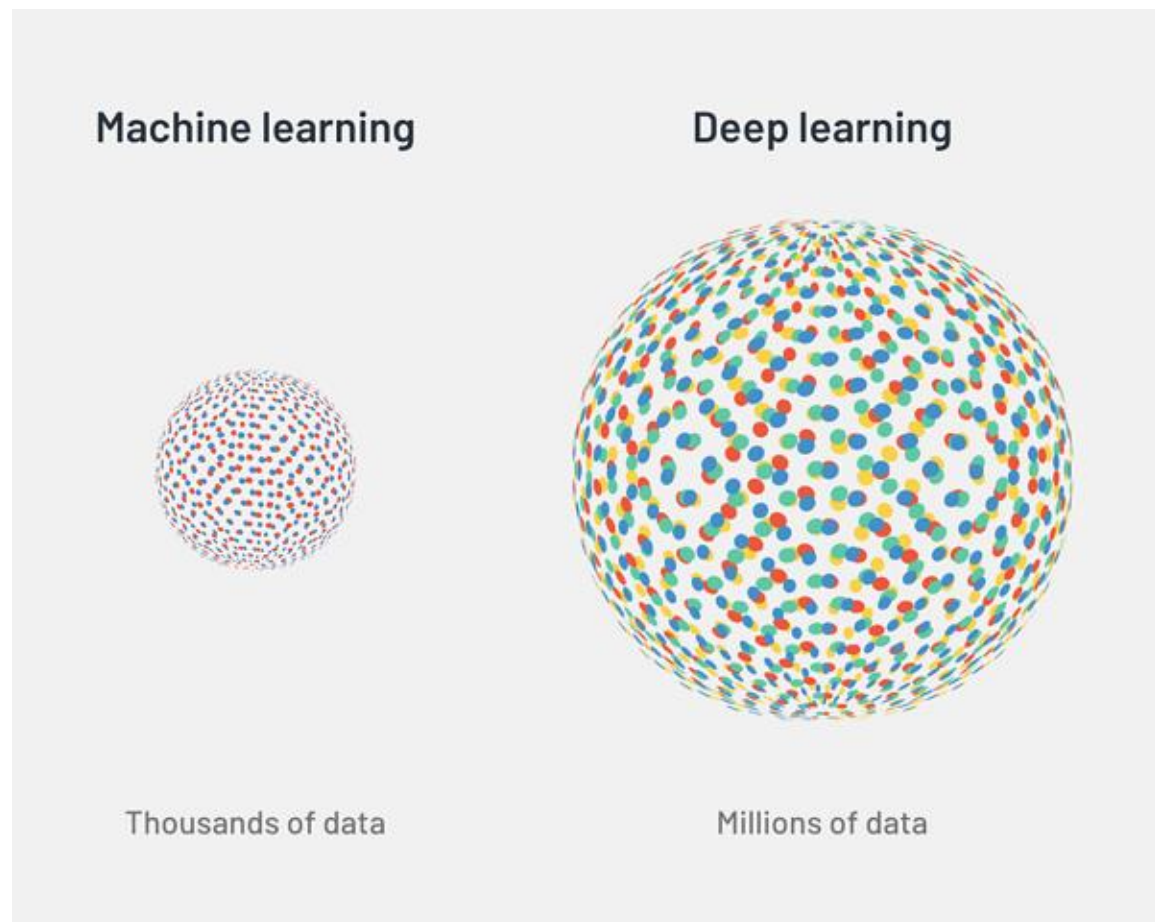
Where are we now?

DeepFake - IDTransfer





Computer Vision: Deep Learning data





Computer Vision: Deep Learning data

Fortunately, data are everywhere!



Google
Image Search

Picasa™

flickr™

webshots

picsearch™

YouTube
Broadcast Yourself™



Computer Vision: Deep Learning data

But it's less easy to find labeled data!

Data

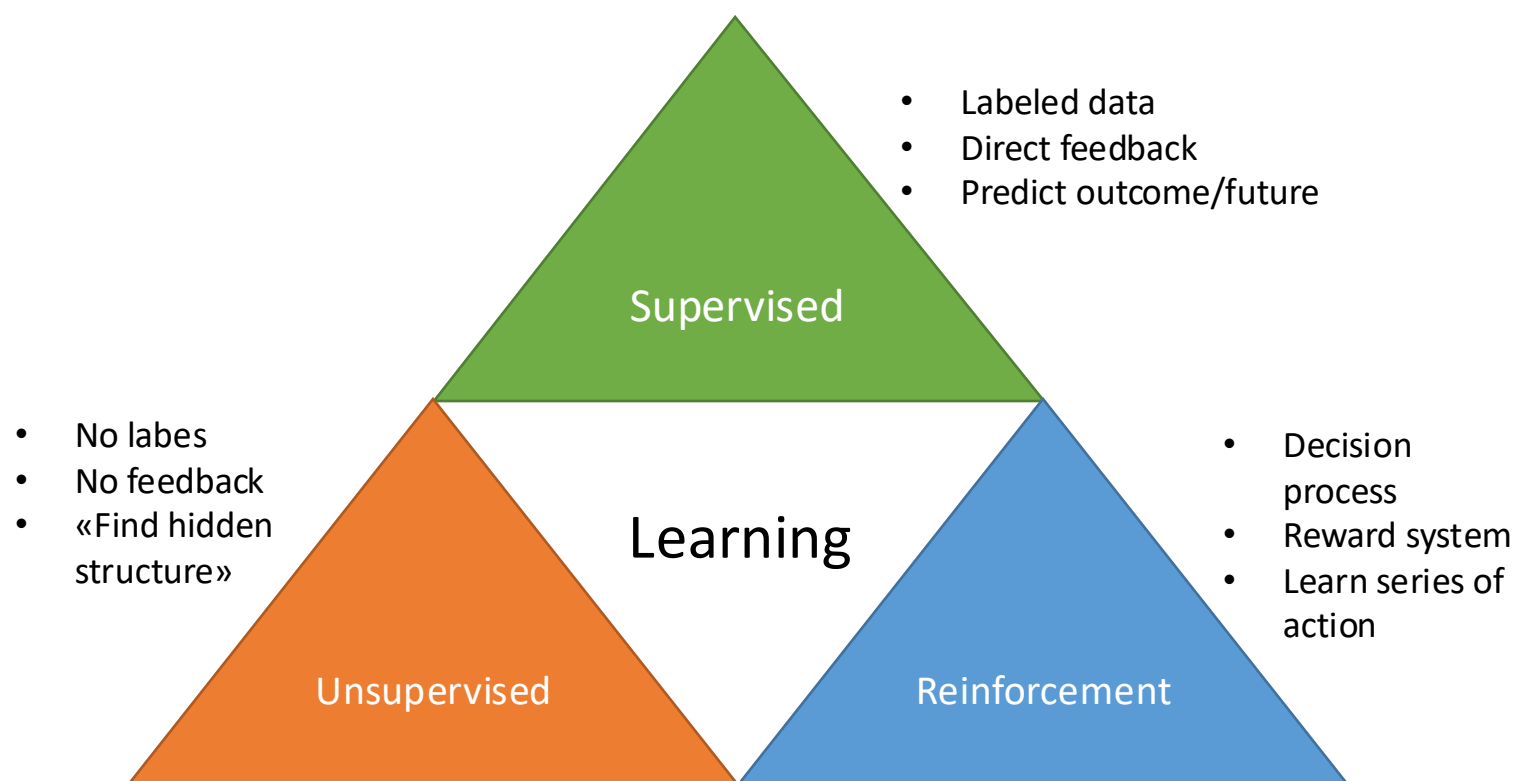


Label

Cat



Not a cat



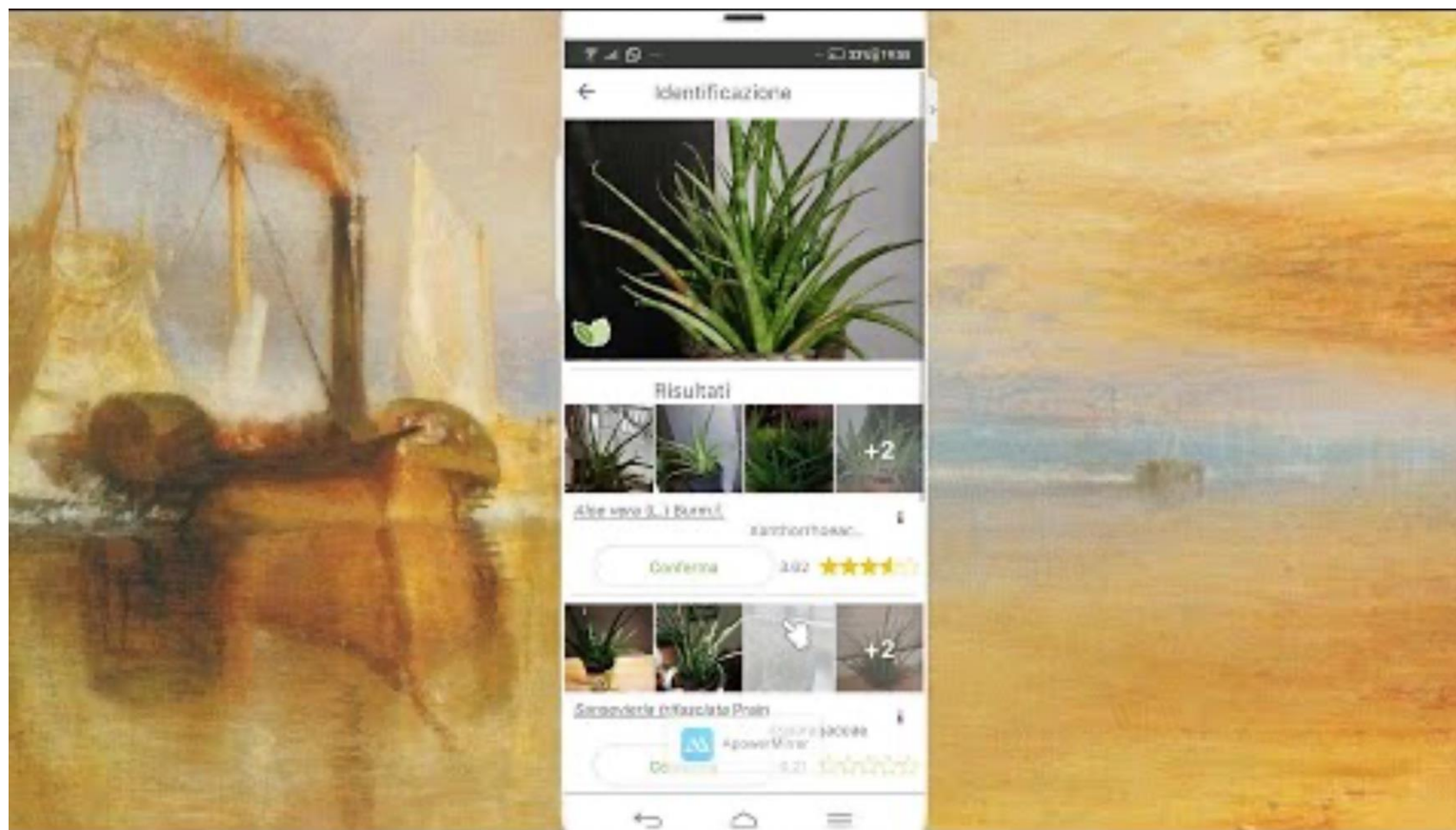


Computer Vision: Combining Deep Learning systems





Computer Vision: Deep Learning Apps





Computer Vision: Deep Learning Apps



Google Lens

Gemini

